

The Bounded Curated Store vs. the Regular-Only Regime: Contract, Selection Rule, and the 2026-07-30/31 Measurement Campaign

local_mixing working notes

2026-07-31

Abstract

The new bounded curated `.frz` store (1.72 GB; ≤ 20 candidates and ≤ 512 decoded value bytes per key) was installed, its runtime contract implemented (per-store value conventions, expansion-first routing), and a two-scale measurement campaign run against the regular-only baselines: 15 arms on the 20k-gate/64-wire sample and 6 arms on the 100k-gate/512-wire slices, all same-seed pairs. Headline: the curated cascade wins on every measured axis at every scale — circuits up to 71% smaller, ancestry transport up to +149%, schedule and slice asymmetries largely erased — with **zero** verification rejections across $\sim 2\text{M}$ curated splices. The splice-economy decomposition shows why: curated conversions are a dedicated growth stroke (58–62% growing, +2-dominated), non-curated conversions a dedicated compression stroke ($\sim 24\%$ shrinking, $\sim 0\%$ growing) — two stores, each doing the job it is built for. A 2×2 interaction check shows the conclusion is robust to (and strengthened by) turning on `-db-advance`, which was off throughout the campaign.

1 The stores

Old (retired): the unbounded curated store — 6.49 GB, 258 files, a pathological 1.19 GB `shard_66` — kept at `frozen_curated_m1_m11.old6g`. Every historical “430k candidates / multi-megabyte values / never reaches its first checkpoint” observation describes this store.

New (installed, verified): `/home/cc/frozen_curated_m1_m11`, 1,720,215,364 bytes; `shard_66.frz` = 6,350,369 B; `tables.bin` `df8ee453...`; `filters.bin` `8dde6635...`. Bounded guarantees: ≤ 20 candidates/key, ≤ 512 decoded value bytes/key, largest encoded bucket 361 B. Reference lookup cost: $3.8\ \mu\text{s}$ curated, $33\ \mu\text{s}$ regular.

The convention fix. The curated store was built under the pre-2ed0222a swapped-controls polynomial key convention, so its values decode with each gate’s two controls swapped relative to native — the never-diagnosed root cause of “every curated candidate fails verification.” The runtime now takes per-store conventions from the environment (`FROZEN_REGULAR_VALUE_CONVENTION=native`, `FROZEN_CURATED_VALUE_CONVENTION=legacy-swapped-controls`), applying a controls-swap at the single decode choke point. Proof at scale: `cur=hits/0` — zero rejections — in every run of the campaign, under mandatory per-splice exhaustive verification.

2 Routing and selection

Commits `f8afdc7b` + `47053b0f`:

- **Expansion (MIX / ANY):** probe CURATED first, forward key only. Apply the mode’s size rule within the curated answer — for MIX: *random among no-larger spellings if any exist, else random among the minimal ones*. On a complete curated miss, fall back to REGULAR (forward + reverse keys) under the same rule. The reverse canonicalization is deferred to the fallback stage.
- **Compression:** REGULAR only, always — including the ssg SAMF-hiding tiers.
- **Tripwire:** a warn-once alarm if any curated value exceeds the bounded contract.

3 Campaign design

Same-seed pairs, only the store policy changed. **20k battery:** the A/B/C schedule family (A phased 200k MIX $\rightarrow$ COMP; B `p_mix` 0.2; C 0.1) $\times$ {no twist, legacy twists 0.002/0.01, g57-v2 twists 0.002/0.01}, 2M moves, exact ancestry. **100k battery:** slices 1 and 2 of the $n=128$ Gray gadget (512 wires) $\times$ `p_mix` {0.20, 0.10, 0.05}, 2.877M moves, sampled ancestry ($K=256$). `-db-advance` was off throughout (both columns — internally consistent; §6 measures the flag’s effect). `FROZEN_FILTER` off for the battery (trajectory-identical; filters only accelerate misses).

4 Results

4.1 20k sample

arm	size old $\rightarrow$ new	anc old $\rightarrow$ new	span old $\rightarrow$ new	eff old $\rightarrow$ new
A no twist	64,333 $\rightarrow$ 24,320	2,649 $\rightarrow$ 6,606	3,253 $\rightarrow$ 7,112	24.9 $\rightarrow$ 61.7
A legacy .002	135,058 $\rightarrow$ 90,770	187 $\rightarrow$ 164	691 $\rightarrow$ 577	16.2 $\rightarrow$ 20.3
A legacy .01	247,382 $\rightarrow$ 227,632	69 $\rightarrow$ 64	393 $\rightarrow$ 327	12.5 $\rightarrow$ 12.9
A g57v2 .002	107,220 $\rightarrow$ 54,678	1,085 $\rightarrow$ 1,673	1,690 $\rightarrow$ 2,121	19.7 $\rightarrow$ 35.9
A g57v2 .01	202,164 $\rightarrow$ 142,886	447 $\rightarrow$ 441	973 $\rightarrow$ 890	14.1 $\rightarrow$ 18.5
B no twist	121,835 $\rightarrow$ 35,213	3,165 $\rightarrow$ 7,527	3,820 $\rightarrow$ 8,159	32.9 $\rightarrow$ 60.2
B legacy .002	243,674 $\rightarrow$ 211,770	350 $\rightarrow$ 350	1,012 $\rightarrow$ 954	21.5 $\rightarrow$ 23.0
B legacy .01	383,924 $\rightarrow$ 373,367	80 $\rightarrow$ 90	451 $\rightarrow$ 443	16.2 $\rightarrow$ 16.4
B g57v2 .002	157,153 $\rightarrow$ 72,929	1,895 $\rightarrow$ 2,602	2,589 $\rightarrow$ 3,275	28.0 $\rightarrow$ 40.0
B g57v2 .01	270,377 $\rightarrow$ 175,873	620 $\rightarrow$ 825	1,198 $\rightarrow$ 1,410	19.7 $\rightarrow$ 23.5
C no twist	48,493 $\rightarrow$ 25,804	7,941 $\rightarrow$ 9,615	8,626 $\rightarrow$ 10,340	55.5 $\rightarrow$ 76.0
C legacy .002	133,665 $\rightarrow$ 120,777	425 $\rightarrow$ 479	1,111 $\rightarrow$ 1,211	31.8 $\rightarrow$ 33.6
C legacy .01	262,115 $\rightarrow$ 259,558	62 $\rightarrow$ 71	369 $\rightarrow$ 399	21.4 $\rightarrow$ 21.3
C g57v2 .002	78,233 $\rightarrow$ 54,031	3,038 $\rightarrow$ 3,339	3,895 $\rightarrow$ 4,098	42.0 $\rightarrow$ 50.6
C g57v2 .01	168,192 $\rightarrow$ 132,681	921 $\rightarrow$ 1,012	1,625 $\rightarrow$ 1,661	26.1 $\rightarrow$ 28.8

Table 1: Finals at 2M moves. Sizes fall everywhere (most where MIX dominates); transport rises with effective work; the A/B/C schedule gap largely collapses (all no-twist baselines land at 24–35k gates, eff 60–76). All prior orderings (twist arms, rates) survive.

run	size old→new	est. anc old→new	cov old→new	ent old→new
s1 p_mix.20	275,431 → 183,160	12,591 → 19,273	.183 → .245	.518 → .612
s2 p_mix.20	284,635 → 177,101	17,901 → 21,330	.254 → .266	.606 → .635
s1 p_mix.10	178,592 → 142,212	22,774 → 27,678	.298 → .339	.651 → .692
s2 p_mix.10	179,136 → 137,094	29,009 → 30,360	.363 → .369	.709 → .719
s1 p_mix.05	136,927 → 125,059	30,031 → 31,562	.383 → .389	.711 → .720
s2 p_mix.05	135,334 → 120,482	33,443 → 33,658	.416 → .743 [≈]	≈

Table 2: Production scale replicates the 20k findings; gains scale with the MIX fraction. The old slice-1-vs-slice-2 asymmetry (12.6k vs. 17.9k anc at p_mix .20) closes under the cascade (19.3k vs. 21.3k).

4.2 100k / 512-wire slices

4.3 Dynamics

Curated share of successful splices is flat over each 20k run ($C \approx 0.17$, $B \approx 0.28$; legacy arms higher via a weak-COMP denominator) and, on the 100k slices, decays from an early store-friendly peak to stable plateaus ($\approx .08/.15/.26$ by p_mix) — the store keeps serving at the same rate deep into the run, and slice-1/slice-2 curves are near-identical.

Grow/shrink balance: the cascade lowers the growing-conversion fraction by 1–6 points and raises the shrinking fraction (B no-twist cumulative $0.147 \rightarrow 0.200$), which no longer decays late-run — COMP stays productive against the smaller steady-state circuit.

4.4 The splice-economy decomposition

A per-store splice histogram (instrumented rerun; bit-identical trajectories verified at both scales) splits the successful-splice (out → in) distribution:

population	n	shrink	equal	grow	shape
(a) old regular-only (20k B)	710k	.147	.690	.164	grow spread +2/ +3/ +4
(b) new total (20k B)	809k	.200	.641	.160	grow at +2 and +6
(c) new, curated only	220k	.087	.334	.579	+2 (30%), +6 (20%)
(d) new, non-curated	589k	.242	.756	.003	−2/ −4/ −6 only
100k: curated only	601k	.079	.305	.616	same shape
100k: non-curated	1.62M	.233	.760	.007	same shape

Table 3: The two-stroke engine: curated conversions are the growth stroke (the store holds longer equivalents by construction — splits of minimal identities — so the “else random minimal” branch dominates, usually at +2), non-curated conversions the compression stroke ($\sim$ zero growth). The old regime forced one store to do both jobs.

5 Robustness to -db-advance

The whole campaign ran with **-db-advance** off (both columns). A 2×2 same-seed interaction check at the strongest-effect point (20k, C schedule, no twists) asks whether the flag changes the conclusion:

cell	size	anc	span	eff
regular, off	48,493	7,941	8,626	55.5
curated, off	25,804	9,615	10,340	76.0
regular, on	66,957	11,950	12,679	39.9
curated, on	27,761	15,637	16,305	60.6

The curated-vs-regular delta *widens* with the flag on (anc $+1,674 \rightarrow +3,686$; size $-22.7k \rightarrow -39.2k$): ballistic transport amplifies whichever splice economy is in place, and the curated economy is better (the regular regime’s growth-heavy splices both spread and accumulate). The flag itself adds +50–63% transport in both regimes — **directive: -db-advance on in every run unless an A/B explicitly needs it off**. The weak-effect bracket (100k slice-1, `p_mix .05`, both advance-on cells) was still running at the time of writing; with the margin above, only a sign reversal there could matter.

6 Conclusions

1. The bounded curated store is correct (zero rejections at two scales under exhaustive verification), fast (its lookups are not a cost factor), and strictly beneficial under the cascade selection rule at every measured operating point.
2. The mechanism is a division of labor: curated supplies re-encoding diversity (growth in small, structured steps), regular supplies compression. Routing each job to its store beats any single-store policy measured.
3. The best measured operating point on the 20k sample is *curated + db-advance*: anc 15,637 at 27.8k gates — $2\times$ the transport of the best regular-only cell at half its size.
4. Caveats: campaign comparisons are same-seed single trajectories (no replication); **-db-advance** was off in the battery (both sides; §5 shows on-state widens the win); campaign-preflight fingerprints still pin the old store and need regeneration.

A Figures

Old (regular-only, dotted) vs new (curated cascade, solid), per schedule

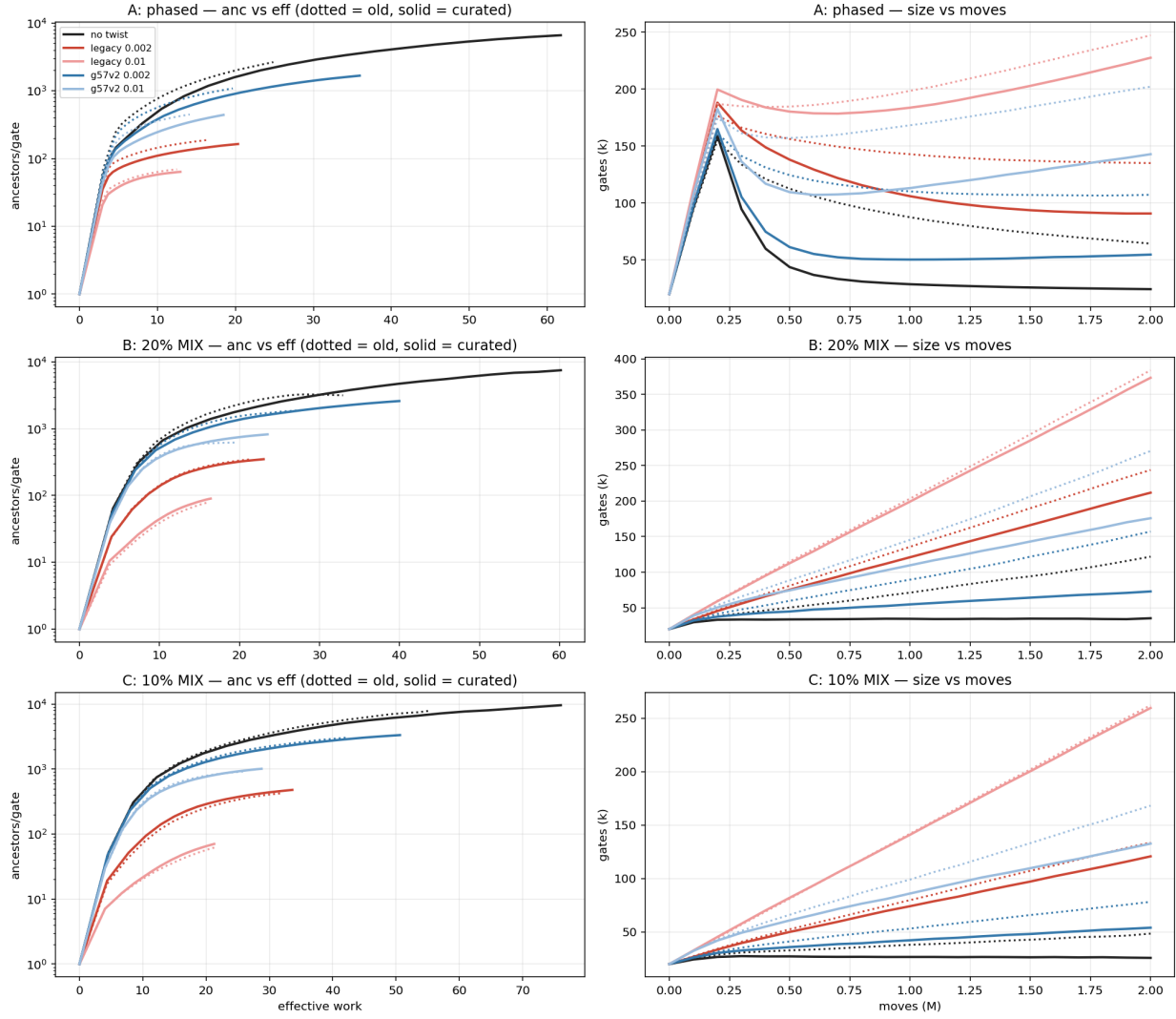


Figure 1: Per-schedule overlays: old (dotted) vs. curated cascade (solid).

100k-gate 512-wire slices — old (regular-only, dotted) vs curated cascade (solid) — sampled ancestry K=256

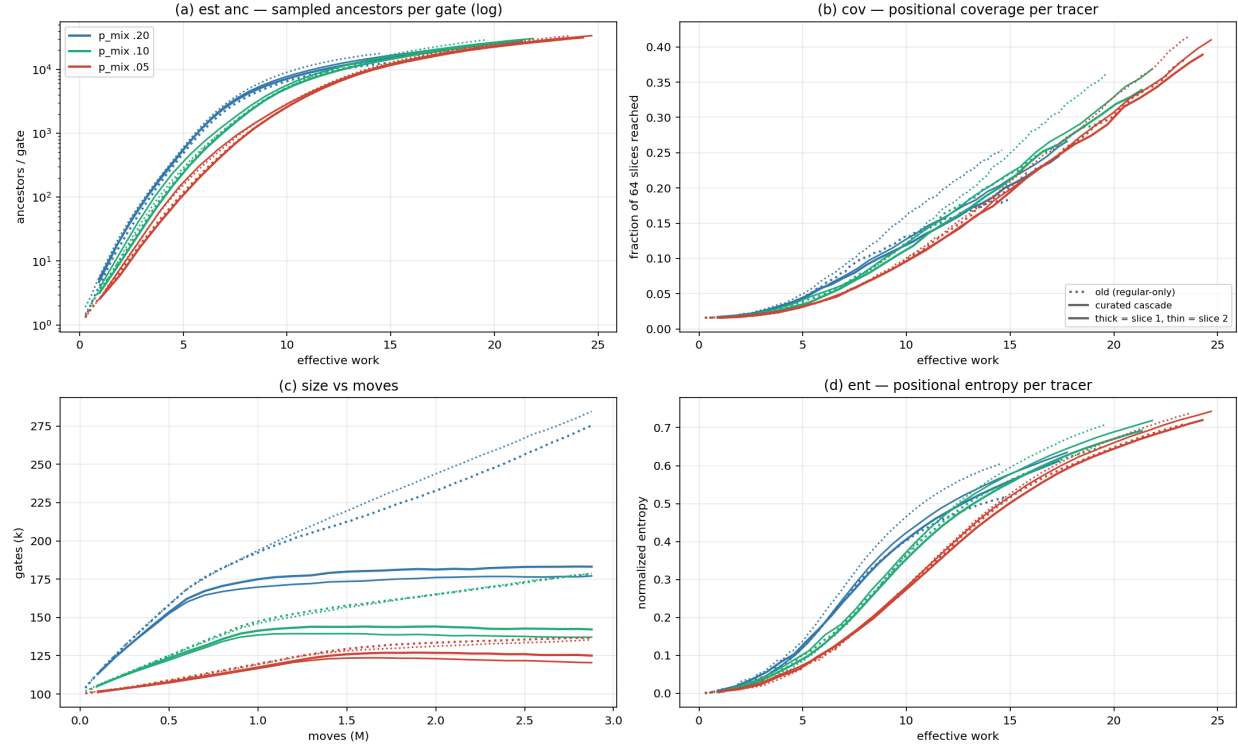


Figure 2: 100k/512-wire slices: sampled-ancestry measures, old vs. new.

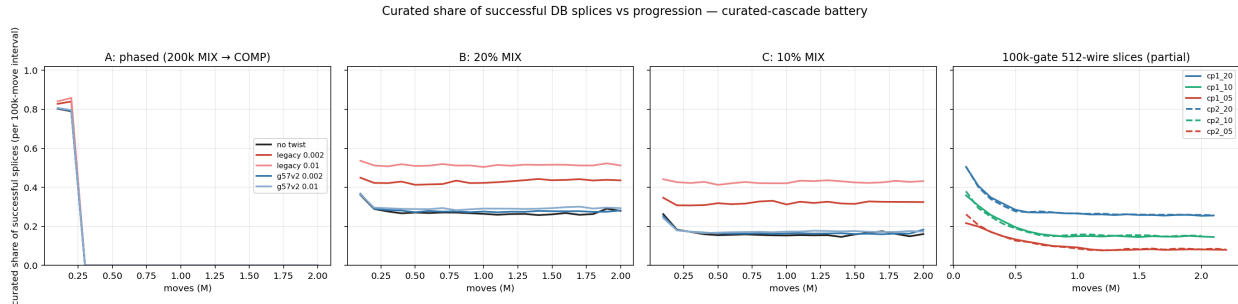


Figure 3: Curated share of successful splices vs. progression.

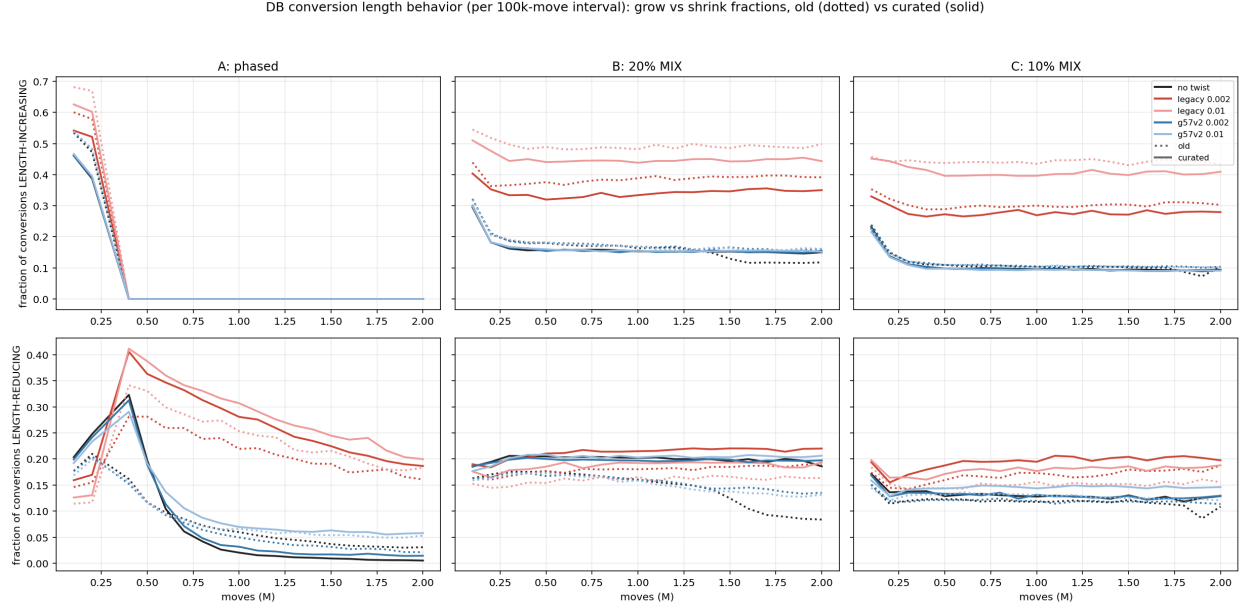


Figure 4: Grow/shrink conversion fractions, old (dotted) vs. new (solid).

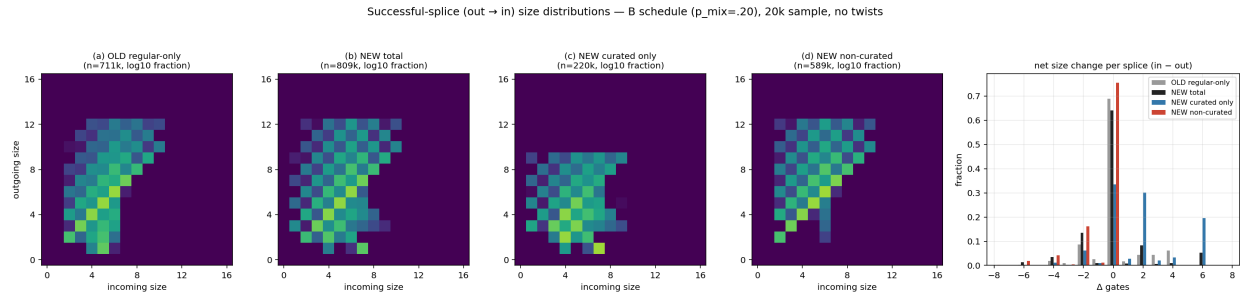


Figure 5: The four-way splice-size decomposition (a/b/c/d).